

Seven Seas A-24

Security Audit

Dec 12, 2024

Version 1.0.0

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Introduction

This document includes the results of the security audit for Seven Seas's smart contract code as found in the section titled 'Source Code'. The security audit was performed by the Macro security team from Dec 9th to Dec 10th, 2024.

The purpose of this audit is to review the source code of certain Seven Seas Solidity contracts, and provide feedback on the design, architecture, and quality of the source code with an emphasis on validating the correctness and security of the software in its entirety.

Disclaimer: While Macro's review is comprehensive and has surfaced some changes that should be made to the source code, this audit should not solely be relied upon for security, as no single audit is guaranteed to catch all possible bugs.

Overall Assessment

No issues or discrepancies were identified within the defined scope of this audit.

Specification

Our understanding of the specification was based on the following sources:

- Discussions with the Seven Seas team.
- Available documentation in the repository.

Trust Assumptions:

Corn's SimpleSwapFacilityPermissioned: Is upgradeable so could potentially update logic and introduce vulnerabilities. Furthermore, the swapExactCollateralForDebt and swapExactDebtForCollateral functions are access-controlled, which means the boring vault must be explicitly granted permission to call them.

Lombard's BTCB: Is upgradeable so could potentially update logic and introduce vulnerabilities.

SonicDepositDecoderAndSanitizer: The contract was verified to match the ABI provided by the Seven Seas team. A more thorough review will be possible once the underlying contracts are provided.

Source Code

The following source code was reviewed during the audit:

- **BTCN Decoder (PR 173) Commit Hash:**

9fb767756890e97154fa350c18e03e4f9432633e

- **Lombard Decoder (PR 174) Commit Hash:**

c4b43f931206a7047d2901dcb08d7a52e0a45cd1

- **Sonic Decoder (PR 12) Commit Hash:**

8a6a85e4f57a128e14c8ed577b4e12ff9d7098fc

- **FluidDex Decoder (PR 166) Commit Hash:**

e4abe13074277d04dd9e9335f808891c8b984153

We audited the BTCN Decoder from [PR 173](#).

Source Code	SHA256
src/base/DecodersAndSanitizers/BTCNFullMinterDecoderAndSanitizer.sol	db60c5cabf082e919c60815edbc8ba4b960911d3d452269d5a376380084b12
src/base/DecodersAndSanitizers/Protocols/BTCNMinterDecoderAndSanitizer.sol	8682611a5e6b50c5b0bbcbdf8311eea1a2866e91dfc8f6416fa9fbf408582be4

The Lombard Decoder from [PR 174](#).

Source Code	SHA256
src/base/DecodersAndSanitizers/Protocols/LombardBtcMinterDecoderAndSanitizer.sol	fc900a73a394f747225c07d9550eee13bac8c0834196f9bea63f3af7f15a6fd5

The Sonic Decoder from [PR 12](#).

Source Code	SHA256
src/base/DecodersAndSanitizers/Protocols/ SonicDepositDecoderAndSanitizer.sol	188bfdf35081d63931b359b6634a064c9 c399be9f0594a04cb3bbae4fad19f5
src/base/DecodersAndSanitizers/SonicFullD epositDecoderAndSanitizer.sol	5b236822f8bf46ca3d7a85696cae1382f8 aaceaca776b8a650f6273c99d21097

The FluidDEX Decoder from [PR 166](#).

Source Code	SHA256
src/base/DecodersAndSanitizers/Protocols/ FluidDexDecoderAndSanitizer.sol	2e2e5f3113f6b52c01098690c754a11182 89987a321f076f242d2c6404de0d76
src/base/DecodersAndSanitizers/FluidDexF ullDecoderAndSanitizer.sol	d69a31b77cd7dcc7800bbd55a9d717b8d1 bad4764383002c55b6c17aeb4acae7

Note: This document contains an audit solely of the Solidity contracts listed above. Specifically, the audit pertains only to the contracts themselves, and does not pertain to any other programs or scripts, including deployment scripts.

Security Level Reference

We quantify issues in three parts:

1. The high/medium/low/spec-breaking **impact** of the issue:

- How bad things can get (for a vulnerability)
- The significance of an improvement (for a code quality issue)
- The amount of gas saved (for a gas optimization)

2. The high/medium/low **likelihood** of the issue:

- How likely is the issue to occur (for a vulnerability)

3. The overall critical/high/medium/low **severity** of the issue.

This third part – the severity level – is a summary of how much consideration the client should give to fixing the issue. We assign severity according to the table of guidelines below:

Severity	Description
(C-x) Critical	We recommend the client must fix the issue, no matter what, because not fixing would mean significant funds/assets WILL be lost.
(H-x) High	We recommend the client must address the issue, no matter what, because not fixing would be very bad, or some funds/assets will be lost, or the code's behavior is against the provided spec.
(M-x) Medium	We recommend the client to seriously consider fixing the issue, as the implications of not fixing the issue are severe enough to impact the project significantly, albeit not in an existential manner.
(L-x) Low	The risk is small, unlikely, or may not be relevant to the project in a meaningful way. Whether or not the project wants to develop a fix is up to the goals and needs of the project.
(Q-x) Code Quality	The issue identified does not pose any obvious risk, but fixing could improve overall code quality, on-chain composability, developer ergonomics, or even certain aspects of protocol design.
(I-x) Informational	Warnings and things to keep in mind when operating the protocol. No immediate action required.
(G-x) Gas Optimizations	The presented optimization suggestion would save an amount of gas significant enough, in our opinion, to be worth the development cost of implementing it.

Issue Descriptions and Recommendations

There were no significant issues found in the code reviewed.

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The scope of this report and review is limited to a review of only the code presented by the Seven Seas team and only the source code Macro notes as being within the scope of Macro's review within this report. This report does not include an audit of the deployment scripts used to deploy the Solidity contracts in the repository corresponding to this audit. Specifically, for the avoidance of doubt, this report does not constitute investment advice, is not intended to be relied upon as investment advice, is not an endorsement of this project or team, and it is not a guarantee as to the absolute security of the project. In this report you may through hypertext or other computer links, gain access to websites operated by persons other than Macro. Such hyperlinks are provided for your reference and convenience only, and are the exclusive responsibility of such websites' owners. You agree that Macro is not responsible for the content or operation of such websites, and that Macro shall have no liability to your or any other person or entity for the use of third party websites. Macro assumes no responsibility for the use of third party software and shall have no liability whatsoever to any person or entity for the accuracy or completeness of any outcome generated by such software.